

MEP 420: Finite Element Method: Practical 5

Cantilever Beam Deflection using Finite Element Method (FEM) and Comparison with Analytical Solution

1 Aim

To determine the deflection of a cantilever beam subjected to (a) a point load at the free end and (b) a uniformly distributed load (UDL), using the Finite Element Method with Hermite beam elements, and to compare the results with analytical solutions from Strength of Materials.

2 Objectives

1. To model a cantilever beam using Euler–Bernoulli beam theory.
2. To assemble the element stiffness matrix for Hermite beam elements.
3. To apply boundary conditions and external loads, and to impose boundary conditions
4. To obtain nodal displacements and reconstruct the beam deflected shape.
5. To compare FEM results (deflections and convergence) with analytical solutions.
6. To study mesh convergence.

3 Theory

For a cantilever beam of length L , flexural rigidity $E I$, subjected to

1. Concentrated load P , the deflection and slope at $x = L$, and elastic curve is given by

$$w(L) = \frac{P L^3}{3 E I}, \quad \theta(L) = \frac{P L^2}{2 E I}, \quad w(x) = \frac{P}{6 E I} x^2 (3L - x), \quad \theta(x) = \frac{P}{2 E I} x (2L - x)$$

2. UDL of intensity q ,

$$w(L) = \frac{q L^4}{8 E I}, \quad \theta(L) = \frac{q L^3}{6 E I}$$
$$w(x) = \frac{q x^2}{24 E I} (6L^2 - 4Lx + x^2), \quad \theta(x) = \frac{q}{2 E I} \left(L^2 x - Lx^2 + \frac{x^3}{3} \right)$$

In case of Beam, the nodal degrees of freedom are w and θ , viz. Transverse deflection and slope. The elemental stiffness matrix of a two-node beam element is:

$$k_e = \frac{EI}{L_e} \begin{bmatrix} 12 & 6L_e & -12 & 6L_e \\ 6L_e & 4L_e^2 & -6L_e & 2L_e^2 \\ -12 & -6L_e & 12 & -6L_e \\ 6L_e & 2L_e^2 & -6L_e & 4L_e^2 \end{bmatrix}$$

For a concentrated load, the global load vector will contain the signed magnitude of the load only at the transverse displacement DOF corresponding to the point of application; all other entries remain zero.

In FEM, there are two ways in which UDL can modeled.

1. Lumped load vector

- (a) Approximate the uniformly distributed load q by converting it into equivalent nodal forces.
- (b) For a beam element of length L_e , the total load is $q L_e$
- (c) Lumped way: Put $q L_e/2$ on each node as transverse load. Note, here we are ignoring the moment at each node due to the UDL.
- (d) The elemental load vector is: $f_e = q [L_e/2, 0, L_e/2, 0]^T$
- (e) As a result, it cannot reproduce the correct bending shape under distributed loading.

2. Consistent Load Vector:

- (a) Derived using the principle of virtual work.
- (b) The elemental load vector is: $f_e = q [L_e/2, L_e^2/12, L_e/2, -L_e^2/12]^T$
- (c) Note, the non-zero entries in the rotational dof ensures that the bending effects are accounted for.

4 Procedure

Assume the following:

1. Define beam geometry and material properties: E , I , L .
2. Discretize the beam into N_e elements of equal length.
3. For each element, form k_e and f_e (if needed).
4. Assemble global stiffness matrix K and load vector F .
5. Apply the kinematic boundary conditions and a concentrated load at the free end.
6. Solve $K u = F$ for nodal displacements.

7. Plot deformed shape and compare with analytical curve. Observe how the elastic curve obtained from FEM converges with the exact solution as the mesh is refined. (Note, when you plot the deformed shape, consider only the deflection dof)
8. With increasing number of elements ($N_e = 2, 4, 8, 16 \dots$) tabulate tip deflection and percentage error.
9. Repeat the procedure for an UDL of intensity q .

Assume the following: $E = 200$ GPa, rectangular cross-section: width = 40 mm, depth = 80mm, $L = 2$ m, Load cases: 1) Point load at free end: $P = 1$ kN, 2) UDL: $q = 1000$ N/m.

5 Results

1. Cantilever with concentrated load P applied at the end
 - (a) Table 1: Tip deflection from FEM, tip deflection from Theory, percentage error, and number of elements.
 - (b) Figure 1: Deformed shape from FEM against the theory.
 - (c) Figure 2: Mesh convergence plot: Tip deflection vs number of elements
2. Cantilever with UDL q :
 - (a) Same as above, once with consistent load vector and once with lumped load vector.
 - (b) Observe and note the number of elements (fineness of mesh) required in case of lumped load vector and consistent load vector for the FEM elastic curve and tip deflection to converge to theory.

6 Post Practical Quiz

1. What is the governing differential equation for Euler–Bernoulli beam bending?
2. What are the degrees of freedom at each node of a Hermite beam element in Euler–Bernoulli theory?
3. Why must we use a consistent load vector for distributed loads instead of lumped nodal loads?
4. What boundary conditions are imposed at the clamped end of a cantilever beam?
5. If u is a column vector of nodal solution (deflection + rotation), how would you extract the vertical displacements at each node? (*Hint: Indexing*)
6. How would you determine the deflection at any point on the beam which is not a node?
7. What is the role of setdiff in the code when applying boundary conditions?